

TECHNICAL MEMORANDUM

PERRY: Design and Open-Water Verification of a LoRa-Activated Vertical Profiling Float

Pressure Equipped Robot for Research and Yield

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ABSTRACT. PERRY is a vertical profiling float: on a LoRa activation signal it runs three profiles to a set depth, logging temperature, pressure, and depth, and transmits the record on resurfacing. In open water it profiled to 4 m, resurfaced, and returned its data. The author owned the programming, the electronics, and the buoyancy design on a team build; the hull sealing work is described in Section 1 and the limits of the verification in Section 5.

1 Pressure Sealing

The hull is a sealed pressure vessel, and the sealing strategy is the core mechanical problem of the craft. Static o-ring seals close the hull. The seal design covered o-ring groove sizing, seal-stack layout, and selection of a waterproof grease for assembly.

The hull is FDM-printed in ABS on a Bambu Lab printer, which leaves a layered, porous surface unsuited to sealing. Printed sealing surfaces were treated by acetone vapor smoothing, using the solvent pass as a conformal surface treatment to close layer lines before seal assembly.

2 Buoyancy Engine

Depth is driven by a buoyancy engine rather than thrusters. A NEMA17 stepper motor, running through a TMC2209 driver under UART control, changes the craft's displacement to move it between the surface and the target depth. The author sized the engine against the craft's buoyancy budget.

3 Electronics and Firmware

An ESP32 runs the mission logic: arming over LoRa, profile sequencing, sensor logging, and data return. Depth and pressure are read from a Blue Robotics Bar02 pressure sensor on the I²C bus. The stepper driver is commanded over UART. All electronics were assembled and soldered on protoboard.

Table 1. System summary.

Subsystem	Component	Interface / process
Controller	ESP32	mission logic, logging, radio
Pressure and depth	Blue Robotics Bar02	I ² C
Depth drive	NEMA17 stepper, TMC2209 driver	UART
Radio	LoRa module	activation and data return
Hull	FDM ABS, Bambu Lab printer	static o-rings, acetone vapor smoothed
Assembly	protoboard, hand soldered	—

4 Open-Water Verification

The craft completed its mission profile in open water: it profiled to 4 m, resurfaced, and delivered its logged data over the radio link.

5 Known Limitations

- Seal integrity is verified to 4 m only. Maximum safe operating depth has not been characterized.
- Effective LoRa range was not measured; the link is verified only at mission distances.
- Craft mass, displacement, and sensor sample rate were not formally recorded.
- Single Bar02 sensor with no redundancy; a sensor fault ends the mission.

6 Role and Attribution

Team project, Crush Depth float team. The author owned the programming, the electronics, and the physics and buoyancy side of the craft, and previously served as Float Team Technical Lead, taking the subteam through LoRa communications, ESP32 development, stepper motors and drivers, and board assembly.